Project 3: Ion Drive Rocket Analysis

Riley J. Kyle*
Missouri University of Science and Technology, Rolla, Missouri, 65409, United States

Dr. David W. Riggins†
Missouri University of Science and Technology, Rolla, Missouri, 65409, United States

The purpose of this project is to

Nomenclature

T	=	temperature $[K]$
P	=	pressure $[Pa]$
ρ	=	density $\left[kg/m^3\right]$
T_t	=	total temperature $[K]$
P_t	=	total pressure $[Pa]$
u	=	velocity $[m/s]$
M	=	mach number
A	=	area $\left[m^2\right]$
F	=	thrust $[N]$
R	=	gas constant $[J/kgK]$
C_p	=	constant pressure specific heat $[J/kgK]$
γ	=	specific heat ratio
$\dot{m}$	=	mass flow rate $[kg/s]$
I_{sp}	=	specific impulse $[s]$
g_0	=	standard gravitational acceleration $\left[9.81m/s^2\right]$

I. Introduction

The purpose of this project temperature and pressure information is shown in table **??**. This data was used when running the solver at each altitude.

*Aerospace Engineering Student, MAE
†Professor of Mechanical and Aerospace Engineering, MAE

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A. Methodology

The following equation can be used to calculate the propellant mass required to perform a maneuver.

$$\frac{\Delta v}{C} = \ln \frac{m_0}{m_0 - m_p} - \int_{T_0}^{T_f} \frac{D}{Cm} dt - \int_{T_0}^{T_f} \frac{g \cos \theta}{C} dt \tag{1}$$

Because the drag term and the gravity can be ignored, this equation can be simplified to the following.

$$\frac{m_0}{m_0 - m_p} = e^{\frac{\Delta v}{C}} \tag{2}$$

C in this equation is the effective exhaust velocity which can be found by using the following.

$$I_{sp} = \frac{C}{g_0} \tag{3}$$

B. Results

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III. Conclusion

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IV. Appendix A